17.3 ARIES and REGULUS: A Unified and Scalable Hardware-Software Co-Designed NPU SoC Family for On-Device and On-Premises Multimodal Inference

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Abstract

We present a scalable NPU architecture, proven in two SoCs (ARIES and REGULUS), designed to efficiently execute diverse AI workloads from vision to large language models. The architecture scales seamlessly across cores, clusters, and chips via a unified programming model. Our holistic hardware–software co-design preserves model accuracy through a bound- and distribution-aware mixed-precision quantization scheme, while efficiently handling complex activation functions via a non-uniform piecewise-linear method. Silicon measurements demonstrate near-linear performance scaling and high power and bandwidth efficiency on both vision and language workloads.

The shift of intelligent compute to edge devices requires neural processing units (NPUs) to efficiently execute a wide spectrum of workloads—from compute-bound convolutional neural networks (CNNs) [1] to memory-intensive transformer-based large language models (LLMs) [2,3]—under strict mobile power and bandwidth constraints. This diversity spans models, scales, and serving scenarios, creating architectural pressure to sustain performance across compute- and memory-bound regimes, efficiently realize complex activation functions such as GELU and SiLU, and meet heterogeneous latency–throughput targets. Quantization is essential to mitigate memory bottlenecks and increase effective bandwidth and capacity, but accuracy is often limited by the resolution–range trade-off, particularly in the presence of outliers; in practice, dequantization overheads on hardware datapaths can erode these gains.

To address these challenges, we present a scalable NPU architecture developed via holistic hardware–software co-design and demonstrated in two systems-on-chip (SoCs)—ARIES and REGULUS. Our approach combines: (1) a bound-aware, distribution-aware mixed-precision quantization scheme that preserves accuracy while maximizing memory efficiency; (2) a non-uniform piecewise-linear (PWL) method that computes complex activation functions with high fidelity and low cost; and (3) a unified programming model that orchestrates these techniques on scalable hardware. Together, these elements deliver high performance and energy efficiency across a broad range of models and applications. Figure 17.3.1 summarizes the challenges and the proposed solutions.

Figure 17.3.2 presents block diagrams of the NPU and the SoC variants—ARIES and REGULUS. A single NPU cluster integrates four NPU cores and a shared 16MiB shared scratchpad memory (SPM). ARIES implements two such clusters for a total of eight cores connected by a hybrid topology Network-on-Chip (NoC). REGULUS contains a single core with a 1MiB shared SPM. Both chips share the same core microarchitecture while targeting area and power optimization at different operating points.

Each NPU core consists of a Tensor Compute Engine (TCE), Tensor Manipulation Engine (TME), Vector Compute Engine (VCE), a DMA Engine, a Control Unit (CU), and a local SPM. The TCE performs matrix multiplication (MATMUL) and convolution. It includes a vector feeder (VF), a matrix feeder (MF), a MAC array, an aggregator (AG), and a partial-sum buffer (PB). The MAC array sustains 4K MACs under a weight-stationary dataflow. The VF streams activations, and the MF streams weights and matrix-structured data. The MF incorporates an on-the-fly dequantization pipeline to reduce memory bandwidth. The AG performs INT30 accumulation with fused requantization and activation and writes results directly to the local SPM without extra data movement. The TME handles common shape/layout transforms. It supports fully unaligned data access via flexible strides and offsets, while a dual logical-bank memory organization hides latency to sustain throughput matching the local SPM bandwidth. The VCE executes elementwise and vector operations including depthwise convolution, average/max pooling, nonlinear functions, top-k, and datatype conversion. In transformer workloads, MATMUL map to the TCE, while softmax, normalization, and top-k map to the VCE, forming an inter-engine pipeline. The DMA supports 3D strides, gather/scatter, prefetch, and double buffering between off-chip DRAM and on-chip SRAM. Three independent units for weight-read (WR), activation-read (AR), and activation-write (AW) maximize concurrency with the TCE/TME/VCE. The CU fetches, decodes, and dispatches a topologically ordered command stream. A combination of compile-time events and a runtime scoreboard resolves data and resource dependencies among DMA/TCE/TME/VCE, preserving topological order and memory consistency.

To minimize accuracy loss while achieving high compression, we adopt a hierarchical mixed-precision quantization approach as illustrated in Fig. 17.3.3. Offline analysis first assigns static mixed-precision at the layer level (Level 1) and the group level (Level 2) according to global statistics. For fine-grained control, each group then selects one of two advanced methods based on distribution characteristics. When outliers are prevalent, elementwise mixed-precision (Level 3-1) decomposes values into low-precision inliers and high-precision extremes to preserve accuracy without distorting quantization ranges. When values cluster non-uniformly, a LUT-based non-uniform quantization (Level 3-2) maps low-bit indices to high-precision values, achieving both compression and expressiveness. Uniform PWL approximation can incur large errors in certain regions. We propose a non-uniform PWL scheme whose segment density adapts to local distribution complexity, implemented with a hierarchical two-level LUT. The L1 stage computes a band index via an affine transform of the input and maps the slope to a band-dependent segment-offset scale. The L2 stage retrieves the PWL coefficients within the selected band. The final output is produced by an affine transform over the segment offset. Concentrating segments in high-curvature regions reduces approximation error at a fixed LUT capacity.

The processor employs a decomposition-aware MATMUL structure to support multiple formats efficiently within a single compute core. A dual scale aligner normalizes partial sums from heterogeneous quantization domains into a common numeric range and leverages a higher-precision previous-scale aligner to minimize normalization loss. Such hardware–software co-design is instrumental in mitigating system-level memory-bandwidth bottlenecks. In particular, quantization substantially reduces the WRDMA time, while the dequantization overhead is overlapped with WRDMA, MAC, and MF.

The NPU core architecture scales uniformly from a single core to multi-core, multi-cluster, and multi-chip systems under a unified programming model. Figure 17.3.4 illustrates the scalability and the performance gains across multi-core, multi-chip systems. At the core level, REGULUS integrates one NPU core. At the cluster level, four local cores and one Global Control Unit (GCU) form a cluster. The GCU provides hardware-level synchronization and performs sharding of weights and activations across tiles. It also orchestrates collective operations such as broadcast, all-gather, and reduce-scatter, supporting parallelization strategies such as tensor, data, and pipeline parallelism. At the multi-cluster level, clusters synchronize via a cluster-signal handshake. At the multi-chip level, diverse parallel execution strategies, as shown in Fig. 17.3.4(a), scale the architecture further.

Under a unified programming model, high utilization is maintained across single-batch/single-model, multi-batch, and multi-model deployments. At the graph level, the compiler partitions workloads to balance compute and memory traffic. Techniques include activation tiling for SPM reuse, output-depth partitioning for fine-grained load balancing, and the selection among tensor, data, and pipeline parallelism per layer. Figure 17.3.4(b, c) illustrates the scalability of ARIES under different execution modes. In throughput-priority (TP) mode, throughput scales almost linearly with the number of cores, while the latency for a single input slightly increases. In contrast, in latency-priority (LP) mode, the throughput scaling is less pronounced, but the latency is effectively reduced as more cores are utilized. This trade-off applies across different batch-size scenarios: LP mode is typically employed in single-batch execution, whereas TP mode is preferable for multi-batch workloads. For multi-model or large models, the compiler applies deterministic partitioning to ensure balanced DMA bandwidth. The same barrier semantics scale from 1 to N cores, clusters, and chips, sustaining near-peak utilization over diverse workloads.

Figure 17.3.5 summarizes the chip-level measurements collected at room temperature under the same software stack. Voltage–frequency shmoo plots are reported for representative vision models, and voltage–performance–efficiency curves show the existence of an operating point with peak energy efficiency. Power breakdown highlights that the NPU is the dominant on-chip consumer during inference, while DRAM, PCIe, and PMIC contribute at the board level.

Figure 17.3.6 compares ARIES to GPUs of similar or higher class across several models. In Fig. 17.3.6(a), GPU baselines are reported in FP16, whereas Fig. 17.3.6(b) uses INT8 baselines. While the GPU can show higher absolute throughput in certain configurations, ARIES demonstrates substantially higher power and bandwidth efficiency. Figure 17.3.6(b) evaluates REGULUS against a representative edge AI platform in a single-batch setting. Despite having lower DRAM bandwidth, REGULUS achieves higher inference speed on key models and better bandwidth efficiency, which is critical in edge and mobile environments.

Tables in Fig. 17.3.6(c) and Fig. 17.3.6(d) indicate that our hardware–software co-designed quantization preserves accuracy close to FP16 on vision tasks and LLMs. For LLMs, we

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evaluate our quantization approach on four language models: Llama-3.2-3B/3.1-8B (Instruct) and Qwen-3-4B (Instruct)/3-8B (base; no public Instruct). Following standard practice, we use WikiText2 as our evaluation dataset, with 128 randomly selected sentences of 2048 tokens for calibration and 100 sentences for evaluation. We report perplexity (PPL) as our primary metric to assess model quality after quantization. Our method employs static quantization for activations, with average bit-widths for both activations and weights detailed in Fig. 17.3.6(d). We build upon GPTQ [4] and SmoothQuant (SMQ) [5] as foundational quantization techniques and enhance them by incorporating elementwise mixed-precision (Level 3-1) and LUT-based non-uniform quantization (Level 3-2) to achieve improved performance while maintaining computational efficiency. The experimental results demonstrate that our method achieves superior PPL scores compared to the baseline GPTQ and SMQ methods, validating the effectiveness of our enhanced quantization strategy. Fig. 17.3.7 shows the die photos of ARIES and REGULUS and summarizes key characteristics of each chip.

In summary, this work presents a scalable NPU architecture, proven in two SoCs (ARIES and REGULUS). The core microarchitecture with TCE/TME/VCE, and efficient mixed-precision support accelerates diverse networks effectively. As the number of cores scales from one to eight, system performance scales nearly linearly. Latency or throughput priority mode enables flexible adaptation to a variety of deployment scenarios.

References:

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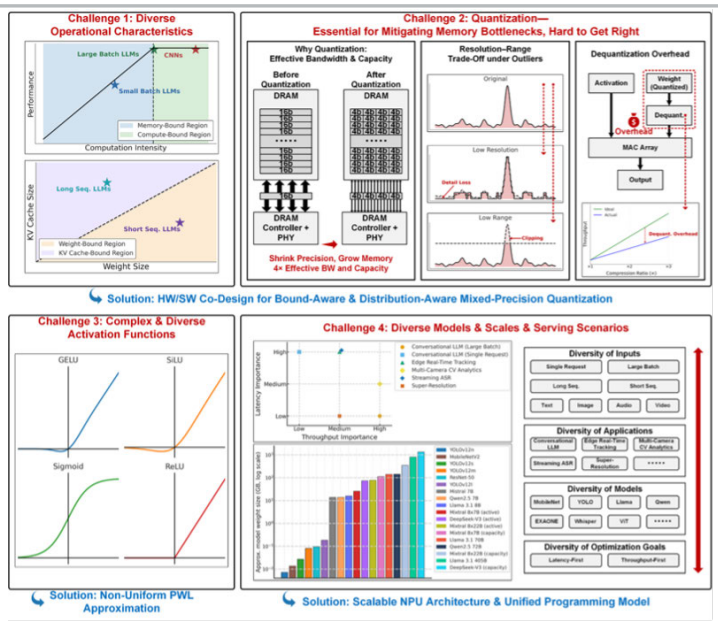

Figure 17.3.1: Challenges and solutions.

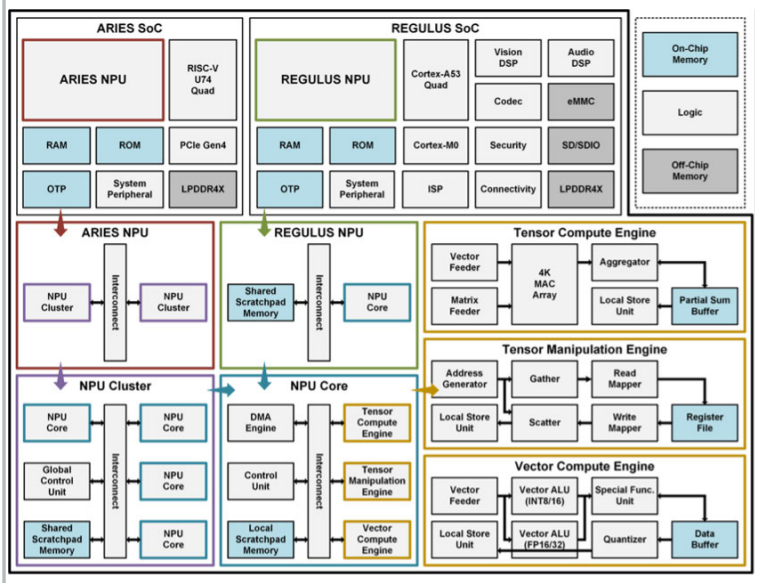

Figure 17.3.2: SoC architecture and NPU microarchitecture diagram.

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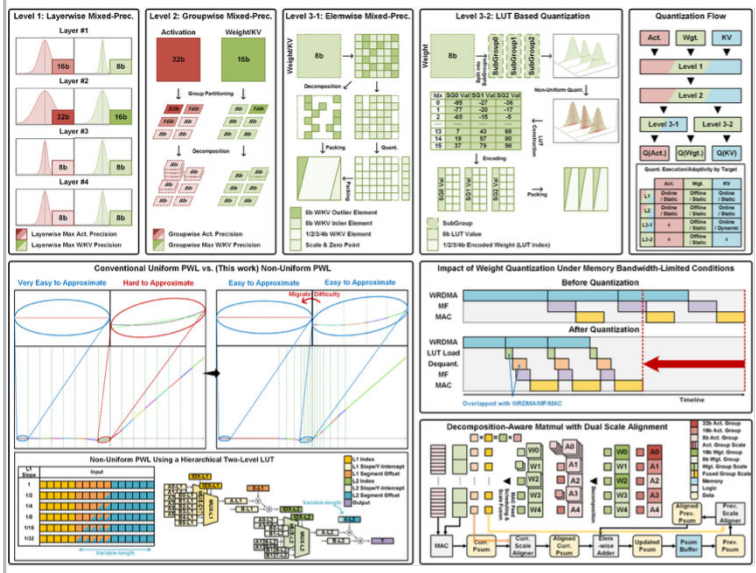


Figure 17.3.3: Mixed-precision execution: INT8 efficiency with FP16 accuracy.

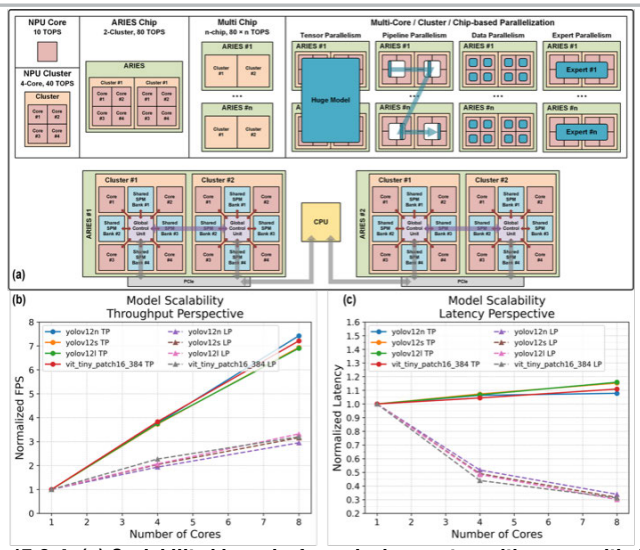


Figure 17.3.4: (a) Scalability hierarchy from single-core to multi-core, multi-cluster, and multi-chip; various parallelism modes on multi-cluster scenario; (b) Normalized FPS scaling in throughput-priority (TP) and latency-priority (LP) modes; (c) normalized scaling in TP and LP modes.

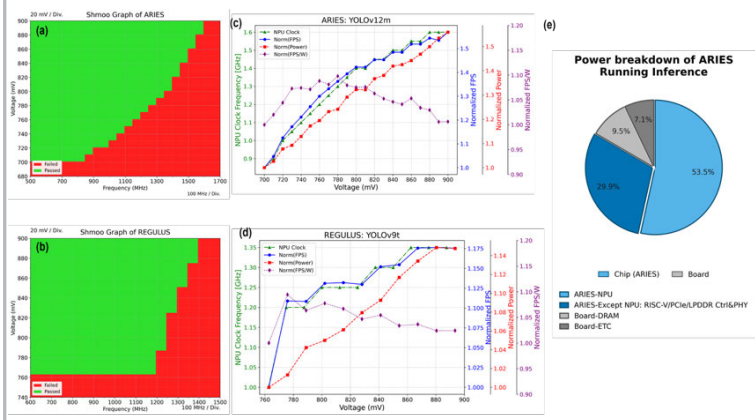


Figure 17.3.5: Room-temperature shmoo plots of (a) ARIES and (b) REGULUS; (c) ARIES running YOLOv12m and (d) REGULUS running YOLOv9t under voltage scaling, showing NPU clock frequency, normalized FPS, normalized power, and normalized efficiency (FPS/W); (e) power breakdown of ARIES during inference.

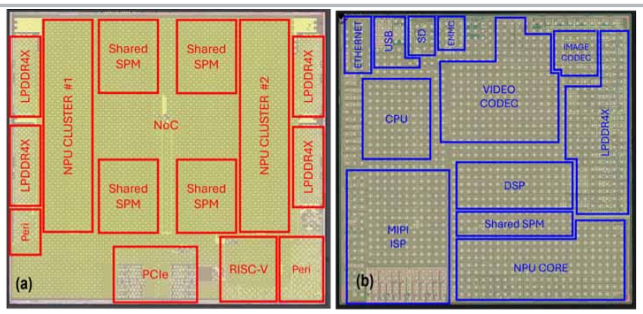
(a) Metrics	FPS(TPS)				FPS(TPS)/W				FPS(TSP)/BW			
Devices	ARIES	[6] RTX 5080	[7] RTX A5000	[8] RTX 2000Ada	ARIES	[6] RTX 5080	[7] RTX A5000	[8] RTX 2000Ada	ARIES	[6] RTX 5080	[7] RTX A5000	[8] RTX 2000Ada
Models \ Device Process	SAM 14N	TSMC 4N	SAM 8N	TSMC 4N	SAM 14N	TSMC 4N	SAM 8N	TSMC 4N	SAM 14N	TSMC 4N	SAM 8N	TSMC 4N
Llama-3.2-3B (B = 1, IL = 64, OL = 64)	19.6	55.49	53.72	31.24	1.04	0.29	0.24	0.49	0.31	0.06	0.07	0.12
MobileNet_v2 (B = 8)	11,599.02	12,981.14	7,766.94	4,638.95	541.23	77.73	38.02	66.84	181.24	13.52	10.11	18.12
YOLOv11s (B = 8)	767.73	1020.51	637.43	372.08	36.06	5.87	3.56	5.97	11.99	1.06	0.83	1.45
YOLOv12n (B = 8)	706.66	1141.11	722.62	391.61	31.20	8.48	3.60	5.64	11.04	1.19	0.94	1.53
YOLOv12m (B = 8)	170.2	559.158	308.60	90.31	6.94	2.78	1.40	1.31	2.66	0.58	0.40	0.35
YOLOv12l (B = 8)	101.71	396.663	216.36	68.53	4.22	1.49	1.00	0.99	1.59	0.41	0.28	0.27
ViT_Tiny_Patch16_224	1707.01	10,040.95	5984.04	2623.81	72.84	68.43	27.77	38.16	26.67	10.46	7.79	10.25

(b) Metrics	FPS		FPS/W		FPS/BW	
Devices	REGULUS	[9] Jetson Orin Nano	REGULUS	[9] Jetson Orin Nano	REGULUS	[9] Jetson Orin Nano
Models \ Device Process	TSMC 12N	SAM 8N	TSMC 12N	SAM 8N	TSMC 12N	SAM 8N
MobileNet_v2	1299.5	760.4	246.64	58.13	152.39	11.15
YOLOv12n	82.42	66.73	14.65	4.49	4.83	0.98
YOLOv12m	24.39	24.99	4.03	1.54	1.43	0.37
YOLOv12l	45.33	14.26	7.24	0.88	2.66	0.21

(c)	Top-1 Accuracy (%) / AP@[0.5:0.95] (%)		
Models \ Methods	Baseline(FP16)	This Work	This Work/FP16
MobileNet_v2	71.81	71.45	99.5
ResNet50	76.27	75.74	99.3
YOLOv11s	44.54	44.32	99.5
YOLOv12s	46.03	45.75	99.4

(d)	WikiText2 PPL (avg. activation bits / avg. weight bits / PPL)			
Models \ Methods	Baseline(FP16)	SMQ+GPTQ (W4)	SMQ+GPTQ (W3)	This Work
Llama3.2-3B-Instruct	16 / 16 / 10.8701	13.93 / 4 / 27.9668	13.93 / 3 / 448.9363	13.93 / 3.25 / **13.6564**
Llama3.1-8B-Instruct	16 / 16 / 7.8234	14.44 / 4 / 13.2731	14.44 / 3 / 1208.7207	14.44 / 3.25 / **9.5106**
Qwen3-4B-Instruct	16 / 16 / 10.7318	14.07 / 4 / 12.5403	14.07 / 3 / 30.8075	14.07 / 3.25 / **12.3732**
Qwen3-8B	16 / 16 / 10.3378	14.28 / 4 / 12.5243	14.28 / 3 / 30.3584	14.28 / 3.25 / **11.6498**

Figure 17.3.6: (a) Performance and efficiency comparison between ARIES and GPUs; (b) performance and efficiency comparison between REGULUS and Jetson Orin Nano; quantization accuracy comparison of (c) CNN models and (d) language models.



(C)	ARIES	REGULUS
Process	SAMSUNG 14nm	TSMC 12nm
Die Area	181.11 ㎟	49.85 ㎟
Clock Frequency	1.25 GHz	1.25 GHz
# of NPU Cores	8	1
TOPS (INT8)	80	10
Precision: W	1, 2, 3, 4, 8 bit	1, 2, 3, 4, 8 bit
Precision: Act	8, 16 bit	8, 16 bit
Shared SRAM	32 MiB	1 MiB
DRAM Capacity	32 GB	8 GB
TDP	25W	5W

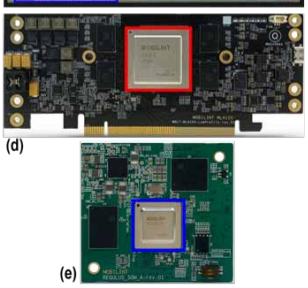


Figure 17.3.7: Die micrograph of (a) ARIES and (b) REGULUS; (c) chip specifications; photographs of the (d) ARIES and (e) REGULUS board and chips.

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